

Open Source Tools and Cryptography

UNIT II and Unit III

Introduction to Antivirus Software

- Antivirus software protects computers from **malware threats**.
- Detects and removes **viruses, worms, and ransomware**.
- Constantly updated to identify **new threats**.
- Uses cryptography to **secure communication**.
- Maintains system integrity by **verifying file authenticity**.
- Ensures **safe system updates** from trusted sources.
- Antivirus works in real-time to **monitor potential threats**.

Antivirus

Signature-based Detection

- Traditional method of detecting malware.
- Compares files to a database of known virus signatures.
- Requires regular updates to remain effective.
- Can miss new, unknown threats.

Heuristic-based Detection

- Detects new, unknown threats by analyzing behavior patterns.
- Can identify previously unseen malware.
- Reduces the reliance on signature updates.
- Risk of false positives.

Antivirus Comparison Table

Parameter	Norton	McAfee	Avast	Kaspersky	Bitdefender
Protection Level	High	High	Medium	High	High
Ease of Use	Medium	High	High	Medium	Medium
Real-Time Scanning	Yes	Yes	Yes	Yes	Yes
Free Version	No	No	Yes	No	No
System Performance Impact	Medium	High	Medium	Medium	Low
Ransomware Protection	Yes	Yes	Yes	Yes	Yes
Malware Detection Rate	High	High	Medium	High	High
Multi-Device Support	Yes	Yes	Yes	Yes	Yes
Price (Annual)	High	Medium	Low	Medium	Medium
Frequent Updates	Yes	Yes	Yes	Yes	Yes

Table: Comparison of Common Antivirus Software

Table Parameters

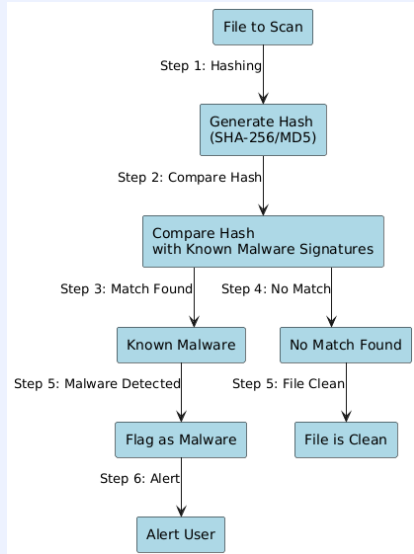
- **Protection Level:** How effective the antivirus is in providing overall protection.
- **Ease of Use:** How user-friendly the interface and usability are.
- **Real-Time Scanning:** Whether the software provides real-time scanning functionality.
- **Free Version:** Availability of a free version.
- **System Performance Impact:** The effect of the antivirus on the system's performance.
- **Ransomware Protection:** Whether the antivirus offers protection against ransomware.
- **Malware Detection Rate:** How effective the software is at detecting malware.
- **Multi-Device Support:** Whether the antivirus supports protection for multiple devices.
- **Price (Annual):** General price range for an annual subscription.
- **Frequent Updates:** Whether the software is updated regularly to counter new threats.

Role of Cryptography in Antivirus Software

- **Data Integrity:** Uses **cryptographic hashes** (e.g., SHA-256, MD5) to verify file integrity.
- **Encryption:** Ensures that antivirus data is **securely transmitted** using **AES or RSA**.
- **Digital Signatures:** Verifies **software authenticity** using cryptographic signatures.
- **Updates:** Ensures **safe delivery** of antivirus updates without tampering.
- **License Validation:** Uses encryption to prevent **unauthorized use** of antivirus software.
- **Encrypted Logs:** Protects **incident reports** and logs from external access.
- **Network Security:** Secures communications between **servers and antivirus clients**.

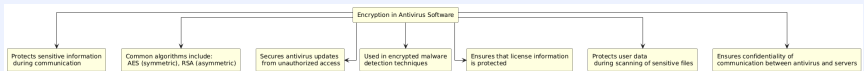
Hashing in Antivirus Software

- Hashing generates a **unique fingerprint** for files.
- Ensures **data integrity** during malware scanning.
- Common hashing algorithms: **SHA-256, MD5**.
- Used for **file signature comparison** to detect malware.
- **Efficient** for detecting known threats.
- Ensures that software updates remain **authentic**.
- Protects against **file tampering** by verifying file hashes.

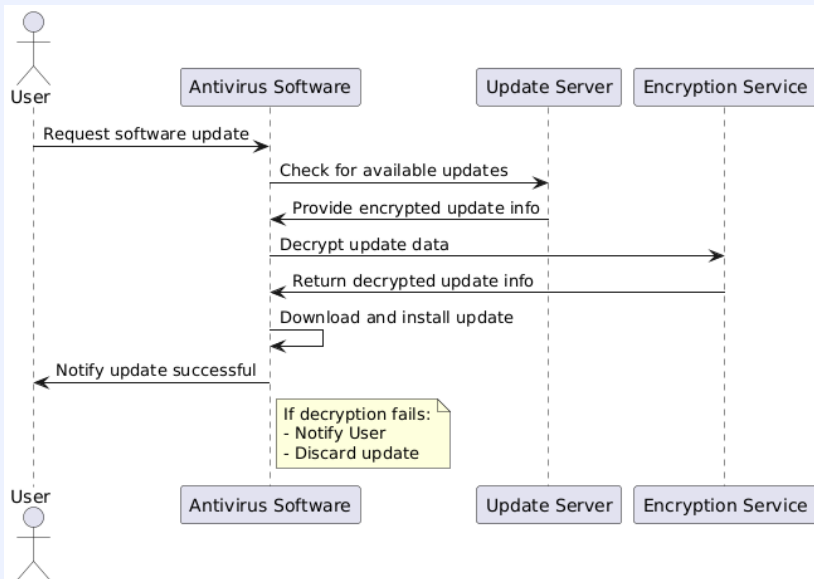


Encryption in Antivirus Software

- Protects sensitive information during **communication**.
- Common algorithms include **AES (symmetric)** and **RSA (asymmetric)**.
- Secures **antivirus updates** from unauthorized access.
- Used in **encrypted malware detection** techniques.
- Ensures that **license information** is protected.
- Protects **user data** during scanning of sensitive files.
- Ensures **confidentiality** of communication between antivirus and servers.

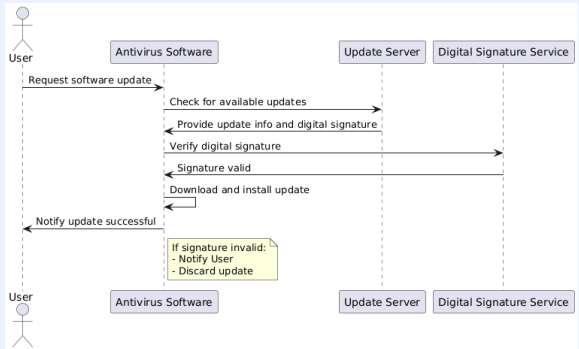


Encryption in Antivirus Software



Digital Signatures in Antivirus Software

- Verifies **authenticity** of antivirus updates.
- Ensures updates are **secure** and **untampered**.
- Uses cryptographic algorithms such as **RSA** or **ECDSA**.
- Ensures that software comes from a **trusted source**.
- Prevents **malicious updates** from being installed.
- Verifies **signatures of scanned files** for enhanced security.
- Protects against **supply chain attacks** on antivirus software.



The McCumber Cube

Security Goals

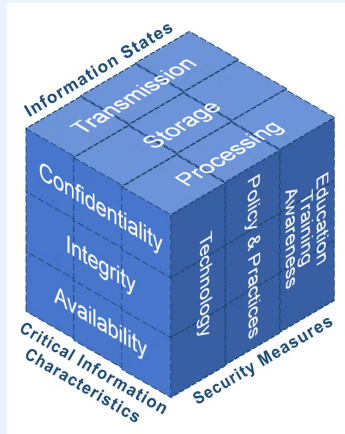
- Confidentiality
- Integrity
- Availability

Information States

- Storage
- Transmission
- Processing

Security Measures

- Policies and Procedures
- Technology
- Education and Training



Description: The McCumber Cube is a 3D framework for assessing and enhancing cybersecurity, covering the intersections between security goals, information states, and measures.

Introduction to Ciphers

- A **cipher** is a method of transforming text to secure information.
- Main categories of ciphers:
 - Substitution Ciphers
 - Transposition Ciphers
 - Symmetric Ciphers
 - Asymmetric Ciphers
- Aim to protect **confidentiality** and ensure **data integrity**.
- Ciphers can be applied in various fields including **communication** and **data storage**.

Substitution Ciphers

- Each letter in the plaintext is replaced with a **substitute**.
- Common types:
 - **Caesar Cipher** - shifts letters by a fixed number.
 - **Monoalphabetic Cipher** - substitutes letters with any other letter.
- Example:

Plaintext: HELLO $\Rightarrow$ Ciphertext: KHOOR(using Caesar with shift 3)

- Vulnerable to **frequency analysis**.
- Suitable for **simple encryption** tasks.
- Requires a shared key for decryption.

Caesar Cipher

Definition: The Caesar Cipher is a substitution cipher where each letter in the plaintext is shifted by a fixed number down or up the alphabet.

Formula:

$$C = (P + K) \mod 26$$

where:

- C is the ciphertext letter.
- P is the plaintext letter.
- K is the key (shift value).
- $\mod 26$ ensures the result wraps around the alphabet.

Example: Encrypt the word HELLO with a shift of $K = 3$.

- H becomes K
- E becomes H
- L becomes O
- L becomes O
- O becomes R

Ciphertext: KH00R

Decryption Example

Decryption: To decrypt, we reverse the shift by subtracting the key.

Formula:

$$P = (C - K) \mod 26$$

Example: Decrypt the ciphertext KH00R with $K = 3$.

- K becomes H
- H becomes E
- 0 becomes L
- 0 becomes L
- R becomes O

Plaintext: HELLO

Monoalphabetic Cipher

Definition: The Monoalphabetic Cipher is a substitution cipher where each letter in the plaintext is substituted with another letter based on a fixed mapping.

Key: A fixed substitution key is used to map each letter of the alphabet to another letter. Example of a random substitution key:

Plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext	Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N	M

Example: Encrypt the word HELLO using the key above.

- H becomes I
- E becomes T
- L becomes S
- L becomes S
- O becomes G

Ciphertext: ITSSG

Decryption Example

Decryption: To decrypt, we reverse the mapping using the same key.

Plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext	Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N	M

Example: Decrypt the ciphertext ITSSG using the same key.

- I becomes H
- T becomes E
- S becomes L
- S becomes L
- G becomes O

Plaintext: HELLO

Other Substitution Ciphers

1. Atbash Cipher:

- Letters of the alphabet are reversed (e.g., A becomes Z, B becomes Y).

2. Vigenère Cipher:

- Polyalphabetic cipher using a keyword to shift letters.

3. Playfair Cipher:

- Encrypts pairs of letters using a 5x5 grid.

4. Affine Cipher:

- Uses the function $E(x) = (ax + b) \bmod 26$ for encryption.

5. Beaufort Cipher:

- Reciprocal cipher, similar to Vigenère but with a different algorithm.

Other Substitution Ciphers (continued)

6. Hill Cipher:

- Uses matrix multiplication to transform blocks of letters.

7. Bazerics Cipher:

- Combines the Polybius square with Vigenère-like shifting.

8. Bacon's Cipher:

- Represents letters using binary patterns of A's and B's.

9. Homophonic Substitution Cipher:

- Each letter maps to multiple possible symbols to disguise letter frequency.

10. ADFGVX Cipher:

- Combines a Polybius square with a transposition cipher.

Atbash Cipher

Definition: The Atbash Cipher is a simple substitution cipher where the alphabet is reversed. A becomes Z, B becomes Y, and so on.

Key: The substitution rule is defined as:

Plaintext	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Example: Encrypt the word HELLO.

- H becomes S
- E becomes V
- L becomes O
- L becomes O
- O becomes L

Ciphertext: SVOOL

Decryption Example

Decryption: The decryption process is the same as encryption since the Atbash Cipher is symmetric.

Example: Decrypt the ciphertext SV00L.

- S becomes H
- V becomes E
- 0 becomes L
- 0 becomes L
- L becomes 0

Plaintext: HELLO

Vigenère Cipher

Definition: The Vigenère cipher is a polyalphabetic substitution cipher that uses a keyword.

Encryption formula: $C_i = (P_i + K_i) \bmod 26$

Decryption formula: $P_i = (C_i - K_i) \bmod 26$

Example:

- Plaintext: HELLO
- Key: KEY
- Ciphertext: RIJVS

Vigenère Cipher Detailed Calculation Table

	H (7)	E (4)	L (11)	L (11)	O (14)
Key	K (10)	E (4)	Y (24)	K (10)	E (4)
Ciphertext Calculation	$(7 + 10) \bmod 26$ = 17	$(4 + 4) \bmod 26$ = 8	$(11 + 24) \bmod 26$ = 9	$(11 + 10) \bmod 26$ = 21	$(14 + 4) \bmod 26$ = 18
Ciphertext Result	R (17)	I (8)	J (9)	V (21)	S (18)

Playfair Cipher

Definition: The Playfair cipher is a digraph substitution cipher using a 5x5 key square.

Encryption Rule:

- Letters in the same row or column are shifted.
- Otherwise, form a rectangle and substitute the corner letters.

Example:

- Key Square (using keyword MONARCHY)
- Plaintext: HELLO
- Ciphertext: CF US MP

Playfair Cipher Example

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

- Encode pairs: HE, LX, LO (add filler if necessary).
- Apply Playfair rules to get ciphertext.

Affine Cipher

Definition: The Affine cipher is a substitution cipher using a linear function.

Encryption formula: $C = (a \times P + b) \bmod 26$

Decryption formula: $P = a^{-1} \times (C - b) \bmod 26$

Example:

- Parameters: $a = 5, b = 8$
- Plaintext: HELLO
- Ciphertext: RCLLA

Affine Cipher Example Calculation

Plaintext	H	E	L	L	O
Calculation	$(5 \times 7 + 8) \bmod 26$	$(5 \times 4 + 8) \bmod 26$	$(5 \times 11 + 8) \bmod 26$	$(5 \times 11 + 8) \bmod 26$	$(5 \times 14 + 8) \bmod 26$
Ciphertext	R	C	L	L	A

Beaufort Cipher

Definition: The Beaufort cipher is a polyalphabetic cipher similar to Vigenère but with a different encryption formula.

Encryption formula: $C_i = (K_i - P_i) \bmod 26$

Decryption: The process is identical to encryption.

Example:

- Plaintext: HELLO
- Key: KEY
- Ciphertext: DYYLP

Beaufort Cipher Example Calculation

Plaintext	H	E	L	L	O
Key (repeated)	K	E	Y	K	E
Calculation	$(10 - 7) \bmod 26$	$(4 - 4) \bmod 26$	$(24 - 11) \bmod 26$	$(10 - 11) \bmod 26$	$(4 - 14) \bmod 26$
Ciphertext	D	Y	Y	L	P

Hill Cipher

Definition: The Hill cipher is a polygraphic substitution cipher that uses linear algebra. It encrypts plaintext using an $n \times n$ matrix (the key) to transform groups of n plaintext letters at once.

Encryption formula:

$$\mathbf{C} = \mathbf{K} \times \mathbf{P} \pmod{26}$$

where:

- $\mathbf{C}$ is the ciphertext vector.
- $\mathbf{K}$ is the key matrix (invertible).
- $\mathbf{P}$ is the plaintext vector.

Decryption formula:

$$\mathbf{P} = \mathbf{K}^{-1} \times \mathbf{C} \pmod{26}$$

where $\mathbf{K}^{-1}$ is the inverse of the key matrix modulo 26.

Key points:

- The matrix must be invertible modulo 26 for decryption to work.
- Works in blocks of letters (usually 2 or 3 letters at a time).

Hill Cipher Example (2x2 Matrix)

Example: Encrypting the message "HELLO" using a 2x2 key matrix.

Step 1: Key Matrix

$$\mathbf{K} = \begin{pmatrix} 6 & 24 \\ 1 & 16 \end{pmatrix}$$

This matrix must be invertible modulo 26.

Step 2: Convert Plaintext to Numeric Values Convert the plaintext letters to numbers using the A-Z correspondence (A=0, B=1, ..., Z=25):

Plaintext "HELLO" = H=7, E=4, L=11, L=11, O=14

Group the plaintext into blocks of 2 letters:

$$\mathbf{P} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}, \begin{pmatrix} 11 \\ 11 \end{pmatrix}, \begin{pmatrix} 14 \\ 0 \end{pmatrix}$$

Step 3: Apply the Hill Cipher Formula

$$\mathbf{C}_1 = \mathbf{K} \times \mathbf{P}_1 \mod 26 = \begin{pmatrix} 6 & 24 \\ 1 & 16 \end{pmatrix} \times \begin{pmatrix} 7 \\ 4 \end{pmatrix} \mod 26$$

$$\mathbf{C}_1 = \begin{pmatrix} 6 \times 7 + 24 \times 4 \\ 1 \times 7 + 16 \times 4 \end{pmatrix} \mod 26 = \begin{pmatrix} 42 + 96 \\ 7 + 64 \end{pmatrix} \mod 26 = \begin{pmatrix} 138 \\ 71 \end{pmatrix} \mod 26$$

$$\mathbf{C}_1 = \begin{pmatrix} 138 \mod 26 \\ 71 \mod 26 \end{pmatrix} = \begin{pmatrix} 8 \\ 19 \end{pmatrix}$$

Corresponding ciphertext letters: I and T.

Hill Cipher Decryption

Step 4: Decrypt the Ciphertext

Step 1: Find the Inverse of the Key Matrix We need to find the inverse of the key matrix modulo 26.

$$\mathbf{K} = \begin{pmatrix} 6 & 24 \\ 1 & 16 \end{pmatrix}$$

The determinant $\det(\mathbf{K}) = (6 \times 16 - 24 \times 1) \bmod 26 = 96 - 24 = 72 \bmod 26 = 20$.
Find the modular inverse of 20 modulo 26 using the Extended Euclidean Algorithm.
The inverse of 20 mod 26 is 4.

Step 2: Compute the Inverse of the Key Matrix Modulo 26

$$\mathbf{K}^{-1} = \frac{1}{\det(\mathbf{K})} \times \begin{pmatrix} 16 & -24 \\ -1 & 6 \end{pmatrix} \bmod 26$$

$$\mathbf{K}^{-1} = 4 \times \begin{pmatrix} 16 & -24 \\ -1 & 6 \end{pmatrix} \bmod 26$$

$$\mathbf{K}^{-1} = \begin{pmatrix} 64 & -96 \\ -4 & 24 \end{pmatrix} \bmod 26 = \begin{pmatrix} 12 & 10 \\ 22 & 22 \end{pmatrix}$$

Hill Cipher Decryption Contd.

Step 3: Decrypt Ciphertext Blocks Using the inverse matrix $\mathbf{K}^{-1}$, apply the decryption formula to the ciphertext blocks to retrieve the plaintext.

For example, for the ciphertext block $\mathbf{C}_1 = \begin{pmatrix} 8 \\ 19 \end{pmatrix}$:

$$\mathbf{P}_1 = \mathbf{K}^{-1} \times \mathbf{C}_1 \pmod{26}$$

Continue similarly for the other ciphertext blocks.

Decryption Process

We are given the inverse key matrix $\mathbf{K}^{-1}$ and a ciphertext block, and we want to retrieve the plaintext using the decryption formula:

$$\mathbf{P} = \mathbf{K}^{-1} \times \mathbf{C} \pmod{26}$$

The inverse key matrix is:

$$\mathbf{K}^{-1} = \begin{pmatrix} 12 & 10 \\ 22 & 22 \end{pmatrix}$$

And the first ciphertext block is:

$$\mathbf{C}_1 = \begin{pmatrix} 8 \\ 19 \end{pmatrix}$$

Decryption Calculation for First Block

We apply the decryption formula:

$$\mathbf{P}_1 = \mathbf{K}^{-1} \times \mathbf{C}_1 \mod 26$$

Substituting the values:

$$\mathbf{P}_1 = \begin{pmatrix} 12 & 10 \\ 22 & 22 \end{pmatrix} \times \begin{pmatrix} 8 \\ 19 \end{pmatrix} \mod 26$$

Performing the matrix multiplication:

$$\mathbf{P}_1 = \begin{pmatrix} 12 \times 8 + 10 \times 19 \\ 22 \times 8 + 22 \times 19 \end{pmatrix} \mod 26$$

$$\mathbf{P}_1 = \begin{pmatrix} 96 + 190 \\ 176 + 418 \end{pmatrix} \mod 26$$

$$\mathbf{P}_1 = \begin{pmatrix} 286 \\ 594 \end{pmatrix} \mod 26$$

Modulo 26 Calculation

Now, reduce the results modulo 26:

$$286 \bmod 26 = 286 - 26 \times 11 = 286 - 286 = 0$$

$$594 \bmod 26 = 594 - 26 \times 22 = 594 - 572 = 22$$

So, the decrypted plaintext vector is:

$$\mathbf{P}_1 = \begin{pmatrix} 0 \\ 22 \end{pmatrix}$$

Convert Numeric Values to Letters

Using the A-Z correspondence (A=0, B=1, ..., Z=25), we get:

$$0 \rightarrow A, \quad 22 \rightarrow W$$

Thus, the first two letters of the plaintext are "AW".

Decryption of Remaining Ciphertext Blocks

The decryption process for subsequent ciphertext blocks follows the same steps. For example, for the second ciphertext block

$C_2 = \begin{pmatrix} 11 \\ 2 \end{pmatrix}$, we would compute:

$$P_2 = K^{-1} \times C_2 \pmod{26}$$

This results in the next pair of plaintext letters, and we continue this process for all remaining ciphertext blocks.

Introduction to Bacon Cipher

The Bacon Cipher was created by Sir Francis Bacon in the early 17th century. It is a substitution cipher that encodes each letter of the alphabet as a sequence of five binary-like characters. These characters can be represented by any two distinct symbols, such as "A" and "B", "X" and "Y", or "0" and "1".

Key Features:

- Uses a binary system to encode the alphabet.
- Each letter is represented by a 5-character sequence.
- The two characters used in encoding are typically distinct, like "A" and "B".

Bacon Cipher Key

The Bacon Cipher assigns a unique five-character sequence to each letter of the alphabet. Here is the standard Bacon Cipher alphabet using "A" and "B":

Letter	Binary Representation
<i>A</i>	<i>AAAAA</i>
<i>B</i>	<i>AAAAB</i>
<i>C</i>	<i>AAABA</i>
<i>D</i>	<i>AAABB</i>
<i>E</i>	<i>AABAA</i>
<i>F</i>	<i>AABAB</i>
<i>G</i>	<i>AABBA</i>
<i>H</i>	<i>AABBB</i>
<i>I/J</i>	<i>ABAAA</i>
<i>K</i>	<i>ABAAB</i>
<i>L</i>	<i>ABABA</i>
<i>M</i>	<i>ABABB</i>

Encrypting with Bacon Cipher

Step 1: Prepare the Plaintext

First, remove spaces and punctuation from the plaintext and convert all letters to uppercase.

Example: Let's encrypt the message "HELLO WORLD". - Remove spaces: "HELLOWORLD" - Convert to uppercase: "HELLOWORLD"

Step 2: Encode Each Letter

Using the Bacon Cipher alphabet: - "H" = AABBB - "E" = AABAA - "L" = ABABA - "L" = ABABA - "O" = ABBAB - "W" = BABAA - "O" = ABBAB - "R" = BAAAA - "L" = ABABA - "D" = AAABB

Step 3: Combine the Sequences

After encoding, the message "HELLOWORLD" is represented as:

AABBB AABAA ABABA ABABA ABBAB BABAA ABBAB BAAAA ABABA AAABB

This sequence of letters can now be hidden or encoded in any form, such as printing it with different fonts or placing it in a text with normal-looking characters.

Decrypting with Bacon Cipher

Step 1: Prepare the Ciphertext

The ciphertext consists of a sequence of characters that represent the encoded message. For example, the ciphertext "AABBB AABAA ABABA ABABA ABBAB BABAA ABBAB BAAAA ABABA AAABB" represents the message "HELLOWORLD".

Step 2: Decode Each Five-Character Sequence

Using the Bacon Cipher alphabet: - AABBB = H - AABAA = E - ABABA = L - ABABA = L - ABBAB = O - BABAA = W - ABBAB = O - BAAAA = R - ABABA = L - AAABB = D

Step 3: Combine the Decoded Letters

The decoded message is "HELLOWORLD".

Example of Encoding and Decoding

Example: Let's encrypt and decrypt the message "HELLO WORLD".

Encryption:

Plaintext: "HELLO WORLD"

After removing spaces and converting to uppercase:

HELLOWORLD

Using the Bacon Cipher:

AABBB AABAA ABABA ABABA ABBAB BABAA ABBAB BAAAA ABABA AAABB

Decryption:

Ciphertext: "AABBB AABAA ABABA ABABA ABBAB BABAA ABBAB BAAAA ABABA AAABB"

Decoding:

H E L L O W O R L D

The decrypted message is "HELLOWORLD".

Key Setup

Step 1: Construct the Key Matrices

The cipher uses two 5x5 grids:

$$\text{Grid1 : } \begin{pmatrix} A & B & C & D & E \\ F & G & H & I/J & K \\ L & M & N & O & P \\ Q & R & S & T & U \\ V & W & X & Y & Z \end{pmatrix}$$

$$\text{Grid2 : } \begin{pmatrix} P & Q & R & S & T \\ U & V & W & X & Y \\ Z & A & B & C & D \\ E & F & G & H & I \\ J & K & L & M & N \end{pmatrix}$$

Note: Letters I and J are combined into one slot in the matrix.

Encrypting with Bazeries Cipher

Step 2: Encrypting the Plaintext

To encrypt a message, follow these steps:

- ① Divide the plaintext into pairs of letters. If a pair is repeated, insert an "X" between the letters (e.g., "LL" becomes "LX").
- ② For each pair of letters, locate their positions in the two grids:
 - If the letters are in the same row, substitute them with the letters immediately to their right (wrap around to the start of the row if needed).
 - If the letters are in the same column, substitute them with the letters immediately below (wrap to the top if needed).
 - If the letters are in different rows and columns, form a rectangle and substitute the letters in the opposite corners of the rectangle.

Example of Encryption

Step 3: Example of Encrypting the Message

Let's encrypt the message "HELLO" using the Bazeries Cipher.

Step 3.1: Prepare the Plaintext

The plaintext "HELLO" is split into pairs of letters: "HE", "LL", "O" becomes "LX".
So, we have:

HE, LL, OX

Step 3.2: Locate in the Grids

Using the two grids: - H is at (2,3) in Grid 1. - E is at (1,5) in Grid 1.

For "HE": - The letters are in different rows and columns. - The letter at the opposite corners of the rectangle are C and I. So, "HE" becomes "CI".

- L is at (3,1) and (4,1) in Grid 1 (same column), so we substitute with the letters below: "L" becomes "M" and "L" becomes "P".

- O is at (3,4) in Grid 1, and X is at (5,3) in Grid 2. They are in different rows and columns, so we use the opposite corners: "O" becomes "C" and "X" becomes "Y".
So the ciphertext for "HELLO" is "CIMPY".

Decrypting with Bazeris Cipher

Step 4: Decrypting the Ciphertext

To decrypt the ciphertext, reverse the encryption process:

- ① For each pair of letters in the ciphertext:
 - If the letters are in the same row, substitute them with the letters immediately to their left.
 - If the letters are in the same column, substitute them with the letters immediately above.
 - If the letters are in different rows and columns, form a rectangle and substitute the letters in the opposite corners.

Using the ciphertext "CIMPY": - C and I are at opposite corners, so they are substituted back to "H" and "E". - M and P are in the same column, so they are substituted back to "L". - C and Y are at opposite corners, so they are substituted back to "O". So the decrypted message is "HELLO".

Transposition Ciphers

- Rearranges the letters of the **plaintext** rather than substituting them.
- Common types:
 - **Rail Fence Cipher** - arranges text in a zigzag pattern.
 - **Columnar Transposition** - writes plaintext into a grid and permutes columns.
- Example:

Plaintext: HELLO WORLD $\Rightarrow$ Ciphertext: HLOOL ELWRD

(using Rail Fence)

- More resistant to **frequency analysis** than substitution ciphers.
- Suitable for **secure communications**.
- The key specifies the order of rearrangement.